Hormuz Buhariwalla

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EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science in Mechanical Engineering — May 2028

Texas A&M University — **College Station, TX**
Bachelor of Science in Mechanical Engineering – GPA: 3.97/4.00 — May 2026

École Nationale Supérieure d'Arts et Métiers – Study Abroad Program — **Aix-en-Provence, France**
Earned 17 credit hours and conducted material science research in an international setting — January 2025 – June 2025

SKILLS

CAD & Simulation: SolidWorks, AutoCAD, Abaqus, MATLAB
Manufacturing & Prototyping: DFMA, 3D Printing, CNC Machining, Lathe Machining, Waterjet, Laser Cutting
Programming & Embedded: Python, Arduino, PID, Dynamic System Modeling

PROJECTS

Graduate Research: Upper-Extremity Motion Tracking — **October 2026 – Present**
Musculoskeletal Biomechanics Lab, Carnegie Mellon University

- Developing an IMU-based upper-extremity tracking system to estimate human joint kinematics using wearable sensing, biomechanical models, and motion-capture validation
- Evaluating sensor-fusion and motion-estimation algorithms to improve the accuracy and robustness of upper-extremity tracking during dynamic movement

Senior Capstone Project: Wheelchair Exercise Retrofit Kit – *Project Manager* — **August 2025 – May 2026**

- Designed and prototyped wheelchair-mounted exercise device delivering 3–60 lbf adjustable resistance while weighing < 5 lbs
- Leveraged commercial off-the-shelf components and custom parts to develop a full-body exercise device costing < $150
- Led the design, fabrication, assembly, and iterative testing of braking, retraction, and mounting subsystems
- Validated critical components and mounting interfaces through static, cyclic, and load testing to verify strength and reliability

Inverted Pendulum – *Independent Project* — **June 2025 – April 2026**

- Derived equations of motion and validated system dynamics via MATLAB simulation before prototyping
- Designed and fabricated a cart–pendulum system using Arduino, DC motor, rotary encoder, and 3D-printed components
- Developed and debugged PID control algorithms in MATLAB and Arduino C++ through iteratively diagnosing software, sensor feedback, motor response, and mechanical behavior to stabilize the pendulum

Invent for the Planet (IFTP) – *3rd Place Overall (51 universities, 250 teams, 1,500 students)* — **January 2025 – April 2025**

- Developed a carbon-capture concept for urban deployment that captures CO_2 and converts it to reusable calcium carbonate

PROFESSIONAL EXPERIENCE

Sustainability Startup — **Dallas, TX**
Engineering Intern — **June 2026 – August 2026**

- Developed an embedded system using Arduino, IR break-beam sensor, OLED display, and I²C communication to provide real-time monitoring and external data-logger integration capabilities
- Designed and prototyped a portable solar power system for multiple low-voltage devices, performing load and battery capacity calculations, component selection, power-distribution design, circuit protection, and outdoor packaging
- Developed 3D-printed components such as removable interfaces and alignment features to improve device maintainability

Eaton Corporation – Electrical Engineering Services & Systems — **Lewiston, ME**
Project Engineering Intern — **May 2024 – August 2024**

- Engineered and fabricated an in-house mobile testing rack holding $70K of electrical equipment, enabling in-shop code deployment and equipment setup, thereby accelerating installation and minimizing expensive on-site hours
- Developed and implemented a Python script to migrate a 30-year-old legacy project repository to a cloud-based repository, eliminating tedious manual searches for drawings and making past designs instantly accessible for reuse
- Migrated 25% of legacy projects (80,000+ files) and authored documentation that ensures a smooth, ongoing transition for future engineers to complete the migration

Dillengr Inc. — **Irving, TX**
Engineering Intern — **May 2023 – July 2023**

- Authored comprehensive work instructions for the manufacturing of power distribution units (PDUs) for a Fortune 500 company, ensuring clarity and usability for production teams